PREVALENCE AND ASSOCIATED FACTORS OF DEPRESSION AMONG ADOLESCENTS IN ADDIN FOUNDATION'S RELIGIOUS SECONDARY SCHOOLS IN IPOH AND KUALA KANGSAR, PERAK

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MUHAMMADBAZLI AMINMOKHSIN BIN AMINUDDIN MASTER OF SCIENCE IN PUBLIC HEALTH 2020

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APPROVAL PAGE

This thesis was submitted to the Universiti Kuala Lumpur and has been accepted as fulfilment of the requirement for the degree of Master of Science in Public Health. The members of the Supervisory committee were as follow:

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DECLARATION

I declare that the work in this thesis was carried out in accordance with the regulations of Universiti Kuala Lumpur. It is original and is the result of my own work, unless otherwise indicated or acknowledge as reference work. This topic has not been submitted to any academic institution or non-academic institution for any other degree or qualification.

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LIST OF ABBREVIATIONS

CDC	Centers for Disease Control and Prevention
CDI	Children Depression Inventory
DASS21	Depression, Anxiety and Stress Scale 21
DOSM	Department of Statistics Malaysia
IPAQ-M	International Physical Activity Questionnaire
MNSR	Malaysia National Suicide Registry
MTQ	Maahad Tahfiz Al-Quran wal Qiraat
NGO	Non-Governmental Organization
PHQ-9	9-item Patient Health Questionnaire
RSES	Rosenberg Self-esteem Scale
US	United States
WHO	World Health Organization
YRBS	Youth Risk Behaviours Surveillance

ABSTRACT

Background: Depression is rising worldwide and thus has become one of the major public health concerns. In Malaysia, previous studies had found that the prevalence of depression among adolescents were between 26.2% to 39.7%. World Health Organization (WHO) has stated that depression can lead to suicide, which is the second leading cause of death in people between 15-29 years old. Therefore, this study was conducted to identify the associated factors and prevalence of depression among religious schools' adolescents in Ipoh and Kuala Kangsar, Perak.
Methodology: A cross-sectional study was done in June 2019. A total of 353 school-going adolescents within the age range of 13-18 years were included into the study. Malay versions of Depression, Anxiety and Stress Scale 21 (DASS21) International Physical Activity Questionnaire (IPAQ-M), Rosenberg Self-esteem Scale (RSES), 9-item Patient Health Questionnaire (PHQ-9), Youth Risk Behaviours Surveillance Questionnaire (YRBS) and sociodemographic questionnaire were used.
Results: The prevalence of depression was 51.3%. After adjustments, significant associated factors were female (adjusted odds ratio (aOR): 2.141; 95% CI 1.303 to 3.516), history of being bullied at school (aOR: 2.346; 95% CI 1.430 to 3.848), low self-esteem level (aOR: 16.291; 95% CI 5.326 to 49.829), and moderate self-esteem level (aOR: 4.220; 95% CI 2.329 to 7.647). The overall prevalence of suicidal ideation was 33.7%, while the prevalence among depressed respondents was 54.1%. Depression was found to be a significant predictor of suicidal ideation (OR 8.490, 95% CI 4.937 – 14.599).
Conclusion: The prevalence of depression in this study was high. Further assessments followed by health education programs were recommended.

ABSTRAK

Latar Belakang: Masalah kemurungan semakin meningkat di seluruh dunia, dan ianya telah menjadi salah satu daripada masalah kesihatan awam yang utama. Di Malaysia, kajian terdahulu mendapati bahawa sebanyak 26.2% hingga 39.7% remaja menghidap masalah kemurungan. Pertubuhan Kesihatan Sedunia (WHO) menyatakan bahawa kemurungan boleh membawa kepada bunuh diri, yang merupakan penyebab kedua utama kematian di kalangan mereka yang berumur 15-29 tahun. Oleh itu, kajian ini telah dijalankan bagi mengenal pasti faktor-faktor yang menyebabkan kemurungan, serta kelazimannya dalam kalangan remaja di sekolah-sekolah agama sekitar Ipoh dan Kuala Kangsar, Perak.

Metodologi: Kajian ini telah dilakukan pada bulan Jun 2019. Seramai 353 orang remaja sekolah dalam lingkungan umur 13-18 tahun telah terlibat sebagai sampel kajian. Soalan soal selidik Skala Kemurungan, Kegelisahan dan Tekanan 21 (DASS21), Aktiviti Fizikal Antarabangsa (IPAQ-M), Skala Harga Diri Rosenberg (RSES), 9 Item Kesihatan Pesakit (PHQ-9), Tingkah Laku Berisiko Remaja (YRBS) dan soal selidik sosiodemografi telah digunapakai di dalam kajian ini.

Hasil Kajian: Kelaziman kemurungan di kalangan responden adalah 51.3%. Faktor-faktor yang signifikan ialah jantina wanita, pernah dibuli di sekolah, dan mempunyai tahap harga diri yang rendah ataupun sederhana. Secara keseluruhannya, kelaziman fikiran ingin membunuh diri di kalangan responden ialah 33.7%. Di kalangan responden yang mengalami kemurungan pula, kelaziman fikiran ingin membunuh diri ialah 54.1%. Kemurungan merupakan faktor yang signifikan dalam membentuk fikiran ingin membunuh diri.

Kesimpulan: Kajian ini mendapati kelaziman kemurungan adalah tinggi. Dicadangkan penilaian lanjut diikuti dengan program pendidikan kesihatan dapat dijalankan kepada responden.

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

The trend of depression is rising worldwide and thus has become one of the major public health concerns. According to the World Health Organization (WHO), depression is regarded as the leading factor that causes ill health and disability worldwide, with about 300 million people suffering from it (WHO, 2017). This is essentially worrying, as it shows an 18% increment as compared to surveillance data collected between 2005 to 2015. When we talk about depression in terms of specific age groups, it is found to be common in adolescents. However, it tends to often be unrecognized and untreated, leading to recurrence and is debilitating to the patient. WHO has stated that depression is one of the causes of suicide, which is the second leading cause of death in people aged between 15-29 years old. It includes a staggering amount of 800,000 deaths due to suicide happening every year (WHO, 2018). Hence, it is important to curb this number from further increasing by addressing the prevalence of depression, so that we can plan the proper preventive strategy for the right target group.

In Malaysia, the source of data for the rate of suicide is not really established due to problems in determining actual causes of death, as it carries a huge stigma to the family as well as legal implications. Due to the discontinuation of Malaysia National Suicide Registry (MNSR), the most recent data on suicide cases were from the suicide registry that was published in 2009. The registry reported that suicide rate was 1.18 per 100,000 Malaysia population. However, it was presumed as being underestimated as the registry was limited to medically certified death, while there was almost half of deaths that were not certified medically. On the other hand, suicidality among Malaysia population were also estimated during the National Morbidity Survey of 2011, where the prevalence of suicidal ideation, plan and attempt were 1.7%, 0.9% and 0.5% respectively. Among the identified risk factors include young age group as well as female gender.

It is worth to note that the prevalence of depression is more in those who came from a family with a low household income and living with a single parent or alone. It shows the relevant association between sociodemographic characteristics and depression (Ling et al., 2018). In a study done in the Kingdom of Saudi Arabia, it was found that 30% of female adolescents aged 15 to 19 years old were having depression (Raheel H., 2015). Researchers in another study also had found that there was significant gender difference, in which female adolescents were having a higher prevalence of depression than male adolescents (Javadi M. et al., 2017).

1.2 PROBLEM STATEMENT

Depression is one of the commonest psychosocial disorders that occur during adolescence stage of life (Levy S., 2019). It is due to the fact that adolescence is a period in which intense learning and development are occurring, as well as a high-risk period for the onset of mental problem (*2017 Children's mental health report*, 2017). Depression in adolescents can lead to a more serious public health issue, which is suicide, if not being identified and managed accordingly. It was reported that suicide is currently the leading cause of death for female adolescents aged 15-19 worldwide ("Anxiety and depression in adolescence," 2017). Therefore, besides assessing the magnitude of depression among adolescents, it is important to explore the risk factors that contribute to this problem, to provide a better insight for its management in the future. For this study, the author will focus on adolescents attending religious-based secondary schools under ADDIN Foundation, to explore the problem of depression pertaining to this population.

1.3 RATIONALE OF STUDY

Adolescence is a stage where an individual starts to become much more independent and slowly go out of the direct control of adults. Therefore, psychological wellbeing is of utmost important besides of physical and spiritual, to provide optimum level of ability in facing various life challenges such as maintaining academic performance, coping with stressful events, and dealing with negative and environments. A better and concise understanding of various predictors of depression among adolescents attending religious schools may contribute to the knowledge on how to curb the problem from further arising. It is hope that through the result of this study, it helps the respective stakeholders like the government, school administrators, public and private organizations in planning of programs pertaining to the mental wellbeing of adolescents.

1.4 RESEARCH QUESTIONS

This study aims to answer the following questions:

i) What is the prevalence of depression among the respondents?

ii) What are the risk factors associated with depression among them?

iii) What is the percentage of depressed respondents who had suicidal ideation?

1.5 OBJECTIVES

1.5.1 General objective

To study the prevalence of depression and its associated factors among adolescents in ADDIN Foundation's religious secondary schools in Ipoh and Kuala Kangsar, Perak

1.5.2 Specific objectives

i. To measure the prevalence of depression among respondents
ii. To describe the sociodemographic characteristics of respondents
iii. To assess the respondents' level of physical activities, self-esteem, history of being bullied, and suicidal ideation status
iv. To study the prevalence of suicidal ideation among depressed respondents
v. To predict which factors associated with depression among respondents

CHAPTER 2

LITERATURE REVIEW

2.1 PREVALENCE OF DEPRESSION

Depression is common worldwide and can be managed effectively with the existing modalities of treatment. However, less than half of the people affected by depression receive treatment (WHO, 2018). The reasons behind include lack of resources and trained healthcare providers, social stigma towards mental disorders, and inaccurate assessment of the problem. The prevalence of depression among adolescents has shown an increasing trend, from about 5% in early adolescence to as high as 20% by the end of that time (Thapar et al., 2012). According to Wartberg et al. (2018), a much lower prevalence of depression was recorded among German adolescents, which was 8.2%. In a study among U.S. adolescents by Mojtabai et al. (2016), the researchers found that the prevalence of depression had increased from 8.7% in 2005, to 11.3% in 2014.

In the Asian region, a systematic review on depression among Iranian adolescents done by Sajjadi et al. (2013) found that the mean prevalence of depression was around 44%, varying between 15% to 72%. In another systematic review, researchers had found that the pooled prevalence of depression among secondary school adolescents in mainland China was 24.3% (Tang & Wong, 2018). Damaiyanti (2016) had conducted a study among adolescents attending high schools in Indonesia, in which the prevalence of depression was found to be as high as 52.7%. As for Malaysia, a study on depressive symptoms among adolescents in Kuching, Malaysia showed 26.2% prevalence (Ling et al., 2018), while Wahab et al. (2013) found 39.7% prevalence of depression among secondary boarding school adolescents in Kuala Lumpur, Malaysia.

2.2 FACTORS ASSOCIATED WITH DEPRESSION

A person may develop depression through a complex interaction of biological, psychological and social elements (WHO, 2018). To determine any single risk factor in isolation is difficult because the interaction between several factors may contribute to an increase in the risk of depression, in a probabilistic way (Thapar et al., 2012). With various factors coming together, an adolescent's life can become unbelievably challenging which if not tackled appropriately might finally end up causing depression. Some of the common contributing factors of depression among adolescents are as follows:

2.2.1 Gender

Gender differences in the development of depression are well established among adult age group, however studies have shown that these differences start to manifest earlier than that (Levine, 2017). It is supported by a study done by Rohde et al. (2009), which stated that adolescence is a high-risk period for the onset of depression in females. In a study among adolescents aged 14 to 19 years, Faravelli et al. (2013) found that females had 2.2 times higher risk of developing depression than males. Meanwhile, Breslau et al., (2017) discovered that more than one-third of female adolescents in the U.S. had experienced their first episode of depression during this phase of their life. This was almost three times the risk as compared to male adolescents.

2.2.2 Parents' education level

A previous study in China had found that low parental education level was significantly correlated with depression in Chinese college students (Zhai et al., 2016). A significant association also was found in a South Korean study, where adolescents who had parents with middle school or lower education level showed the highest response rates of depression compared to those with higher education level parents (Park et al., 2012). However, Mohammadi et al. (2019) did not observe parental education level as a significant predictor for depression in adolescents, while Ling et al. (2018) only found mother's education level as having a significant association with depression.

2.2.3 Parents' occupation

Several studies have shown mixed results for association between parental employment and adolescent depression. Between 2000 to 2011, a repeated cross-sectional survey in Finland found that occurrence of severe depression almost doubled among adolescents with unemployed parents (Torikka et al., 2014). On the other hand, Mohammadi et al. (2019) had found that only adolescents with fathers who were unemployed were having an increased risk of depression, whereas having mothers who were housewives gave a protective effect against it. Meanwhile, a recent study by Moeini et al. (2019) among Iranian female adolescents found no statistically significant correlation at all between both parents' occupation and depression.

2.2.4 Parents' marital status

There were several studies that have been conducted to assess the relationship between marital status of parents with depression. In Uganda, Nalugya-Sserunjogi et al. (2016) found that adolescents with a single parent had almost 2 times the risk of being depressed as compared to those with both parents. A study among adolescents in Nigeria by Oderinde et al. (2018) supported the result, where those who came from divorced families were more likely to have depression than their counterparts from intact families. The authors further postulated that the reasons might be because such individuals tend to have a lesser close relationship with their parents, which eventually will lead to depression. Even worse, adolescents from divorced parents who were experiencing depression, have more frequent suicidal thoughts than those from intact families (Hadžikapetanović et al., 2017).

2.2.5 Physical activity

Involvement in regular physical activity has been proven to be beneficial in cognitive, emotional as well as motor functions. Not only it can reduce distress and negative affect in adolescents, it can also exert preventative action towards depression, through facilitating psychological wellness (Archer et al., 2014). In a study among European adolescents, researchers found that physically active individuals were having lower depression levels as well as higher wellbeing as compared to those who were least and less active (Mcmahon et al., 2016). This is congruent with a study done by Bélair et al. (2018), which stated that physically inactive adolescents had about 1.5 times higher risk of developing moderate and severe symptoms of depression, relative to physically active counterparts. However, a different result was obtained in a study conducted by Tajik et al. (2017), whereby there were no differences between different physical activity levels and depression.

2.2.6 Being bullied

Cambridge online dictionary defines bullying as the behavior of a person who hurts or frightens someone smaller or less powerful, often forcing that person to do something they do not want to do. Kim et al. (2017) further stated that adolescents who are different in terms of appearance or behavior from their peers, are more likely to be bully victims. This include cancer survivors or other chronic diseases sufferers that may experience body image changes due to various reasons like side effects of treatment. A study in Netherlands by Garnefski and Kraaij (2014) found that there was a strong correlation between bully victimization and development of depressive symptoms among secondary school students. This is in line with another research, in which Kaur et al. (2014) discovered significant association between being bullied and development of depression among school-going adolescents in Malaysia, with almost 2 times the risk compared to those who were not bullied. However, a different study result was obtained by Ibrahim et al. (2017), in which being a bully victim was not a significant predictor of depression among adolescents.

2.2.7 Self-esteem level

Self-esteem can be defined as an attitude, made up through self-evaluation, which is based on positive and negative aspects of oneself (V. Paz et al., 2016). Ibrahim et al. (2017) stated in their study among adolescents, that having low self-esteem was a significant associated factor of depression, with almost 3 times the risk relative to high self-esteem adolescents. The result supported the findings of an earlier study by Tripković et al. (2015). When we talk about self-esteem of adolescents, bear in mind that it can produce effects in much later stages of life. In a 23-year cohort study carried out in Switzerland, Grue et al. (2014) demonstrated that adolescents whose self-esteem declined during adolescent years, were more likely to develop depression during adulthood stage two decades later.

2.3 SUICIDAL IDEATION

Suicidal ideation is defined by the US Centers for Disease Control and Prevention (CDC) as "thinking about, considering, or planning suicide" (CDC, 2015). In a study by Ibrahim et al. (2017), the researchers found the suicidal ideation prevalence among adolescents aged 13 to 17 years was 27.9%. There were other studies had been carried out among Malaysian adolescents, in which Chan et al. (2016), Ahmad et al. (2014), and Chen et al. (2005) found 6.2%, 7.9%, and 7.0% prevalence of suicidal ideation respectively. In term of age of completed suicide, the National Suicide Registry Malaysia had reported the youngest to be a 12 years old adolescent. This could suggest that suicidal ideation, which precedes an actual suicide, may start even at an early adolescence age.

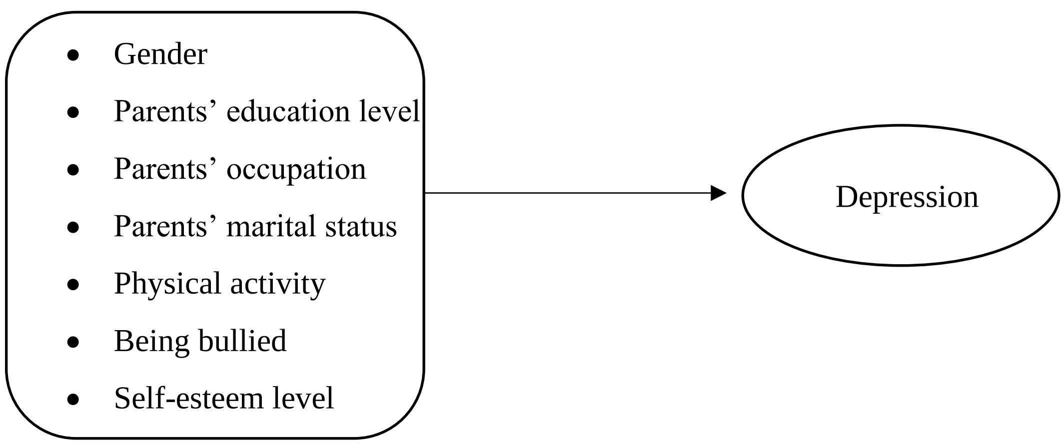

Figure 1: Conceptual framework of associated factors of depression

CHAPTER 3

METHODOLOGY

4.1 OVERVIEW

The study was conducted in five secondary religious boarding schools located in Ipoh and Kuala Kangsar, Perak. The schools are all privately managed by ADDIN Foundation, a non-governmental organization (NGO) in the state of Perak that caters over 20 religious-based secondary schools in Perak and Selangor. According to the Foundation's official website, the NGO uses their own designated syllabus for the students with focus on Tahfiz education (memorization of Al-Quran) and religious subjects (such as *tauhid, akhlaq, and hadith*) as their core syllabus, as well as all mainstream subjects that are compulsory for *Sijil Pelajaran Malaysia (SPM)*, which are Bahasa Malaysia, English, History, Science and Mathematics (Maahad Tahfiz ADDIN, 2010). Three of the schools are located in Ipoh while two are in Kuala Kangsar. The name of the schools are as follows:

1. Maahad Tahfiz Al-Quran wal Qiraat (MTQ) ADDIN 7, Tambun
2. MTQ ADDIN 10, Kampung Sungai Kati, Manjoi
3. MTQ ADDIN 11, Persekutuan Seruan Islam Perak
4. MTQ ADDIN 6, Padang Rengas
5. MTQ ADDIN 13, Kampung Buaia, Kuala Kangsar

4.2 STUDY AREA

<u>Population</u>

Two of the schools, namely MTQ ADDIN 7, Tambun and MTQ ADDIN 10, Kg. Sg. Kati are all-male schools. While MTQ ADDIN 11 Seruan Islam, MTQ ADDIN 6, Padang Rengas as well as MTQ ADDIN 13, Kg. Buaia are all-female schools. The study population were Form 1 until Form 5 students, and all were present in the school compound during the study period. The total number of all students are 353, with 133 males and 220 females. With regards to race, religion, and socioeconomic status, most of them are Malays, all are Muslims, and they are from various background, from low to high income families.

As mentioned earlier, all schools followed a special syllabus, designed by ADDIN Foundation. The students are required to attend morning Quranic memorization classes from 7.30 am to 10.30 am every Monday to Friday, followed by classes on Islamic subjects until 12.30 pm. Then they will have a short resting period for lunch and *Zuhur* prayer until 2.00 pm, when the next classes will start and end at around 4.30 pm. From 4.30 onwards they play sports or do their own leisure activities. At night, they will sit in groups from 8.00 pm until 10.00 pm, to memorize new Quranic sentences to be recited in front of the teachers on the next day, as well as revising previously memorized Quranic sentences. On weekends, they attend classes that focus on SPM subjects from 8.00 am to 12.30 pm. The remaining hours are for them to have their rest and other activities like recreation and revision.

4.3 STUDY DESIGN

A cross-sectional study was conducted among the students of MTQ ADDIN 7, Tambun; MTQ ADDIN 10, Kg. Sg. Kati; MTQ ADDIN 11, Seruan Islam; MTQ ADDIN 6, Padang Rengas; and MTQ ADDIN 13, Kampung Buaia.

4.4 INCLUSION AND EXCLUSION CRITERIA

Inclusion criteria:

1. Adolescents
2. Students of the respective schools who have signed consent forms from their parents

Exclusion criteria:

1. Students who were absent during data collection
2. Students of parents who did not sign the consent form

4.5 SAMPLE SIZE

Sample size required was 338 (with 95% confidence level), calculated using the Openepi online sample size calculator.

Sample Size for Frequency in a Population

Population size(for finite population correction factor or fpc)(*N*):	428600
Hypothesized % frequency of outcome factor in the population (*p*):	32.7%+/-5
Confidence limits as % of 100(absolute +/- %)(*d*):	5%
Design effect (for cluster surveys-*DEFF*):	1

Sample Size(*n*) for Various Confidence Levels

ConfidenceLevel(%)	Sample Size
95%	338
80%	145
90%	239
97%	415
99%	584
99.9%	952
99.99%	1329

Equation

Sample size $n = [DEFF*Np(1-p)]/ [(d^2/Z^2_{1-\alpha/2}*(N-1)+p*(1-p)]$

Results from OpenEpi, Version 3, open source calculator--SSPropor
Print from the browser with ctrl-P
or select text to copy and paste to other programs.

Figure 2: Calculation of sample size by openepi.com (2019)

The sample size for this study was calculated using Open Source Epidemiologic Statistics for Public Health, Version 3.01 at www.openepi.com, updated on 6th April 2013 and accessed on 10th June 2019. According to the latest available data by Department of Statistics Malaysia (DOSM) at this point of study, the total adolescence population of Perak was 428,600 (DOSM, 2018). The absolute precision was 5% and the anticipated prevalence of depression was 32.7% based on a previous study that was done among school-going adolescents by Ibrahim et al. (2017), thus the sample size for this study was

338. As the total number of students were only 353, all students were taken as respondents for this study.

4.6 DATA COLLECTION METHOD AND TOOLS

The data was collected by using self-administered questionnaires, with the researcher present at the time the questionnaire was distributed to answer any queries from the students on the questions. The questionnaires were comprised of the background characteristics of respondents, the validated Malay version of International Physical Activity Questionnaire (IPAQ-M)(Chu & Moy, 2012), validated Malay version of Depression, Anxiety and Stress Scale 21 (DASS21) (Musa et al., 2011), validated Malay version of Rosenberg Self-esteem Scale (RSES) (Yaacob, 2006), validated Malay version of 9-item Patient Health Questionnaire (PHQ-9) (Sherina et al., 2012), and Youth Risk Behaviours Surveillance Questionnaire (YRBS) by Centers for Disease Control and Prevention (2019) (Lee et al., 2007). IPAQ-M was used to assess physical activity status; DASS21 was primarily for assessing degree of depression among respondents; RSES was for identifying self-esteem level; PHQ-9 was for assessing depression as well as suicidal ideation, while YRBS was used for assessing bullying among respondents.

A pre-test was carried out among 30 students from the target population before the actual data collection day. It was meant to detect any difficulty in answering or understanding the questionnaires, and to make modification and improvements in terms of wordings, to make the questionnaires understandable.

4.7 STATISTICAL ANALYSIS

Frequency and percentage (%) were used to describe categorical variables. Association between independent variables and depression was calculated through chi square test, as all of them were converted into categorical variables. After considering variables which had statistically significant association (P-value<0.05), binary logistic regression was done to find the adjusted odds ratio with 95% confidence interval as significant association indicator.

4.8 ETHICAL CONSIDERATION

Prior to data collection, informed consent was taken from each respondent's parents. The researcher had made the effort to ensure that all respondents were briefed and given adequate information on the topic and data collection procedures, before they agreed to participate. They had been informed that all personal information will be kept confidential and private. They were also guaranteed that there are no risks should they participate in this study and can withdraw whenever they felt uncomfortable to continue. The author also had ensured that there was a chaperone during the data collection of female respondents.

Each of the respondent was given a handout on symptoms of depression and where to seek relevant help, such as Befrienders Malaysia organization and Malaysia Suicide Hotlines. After the results of study have been obtained, a meeting was held with the school administrators to arrange a health awareness talk for all students, focusing on depression issue among adolescents. As the school administrators mentioned that they had already planned the school activities for this whole year, they had suggested the health talk to be carried out next year.

The talk is planned to be conducted by UniKL RCMP psychiatrist, Assoc. Prof. Dr. Mohammad Bin Abdul Rahman, together with the researcher. It shall incorporate a 'reach in, reach out' method, whereby the students shall be given necessary input on depression (such as common symptoms and ways to tackle the problem), as well as encouragement to reach out to the researcher and other appropriate individuals like mental health professionals (psychiatrist, psychologist, or councilor) if they ever experience depression. The school administrators and teachers shall also be encouraged to report any student who they believe is having depression (due to reasons such as poor study performance, lack of motivation) to the researcher. Those students who reach out to the researcher shall be further interviewed by the psychiatrist using appropriate diagnostic tools like DSM-V criteria of depressive disorders, to confirm their diagnosis. Their parents will also be informed about their children's condition, and further advised for appropriate intervention by the psychiatrist. The diagnosis of students who reached out to the researcher will be made confidential.

CHAPTER 4

RESULTS

5.1 DEMOGRAPHIC CHARACTERISTICS OF RESPONDENTS

The demographic characteristics of students who participated in this study were shown in Table 1. From the total of 353 respondents, the 37.7% are males while 62.3%, or almost two thirds are females. Based on the forms they were currently in, majority are lower secondary form students.

Regarding parents' level of education, most of the respondents' parents had some level of education. More than half had completed college or university education (185 or 52.4% of fathers; 198 or 56.1% of mothers). Among respondents' fathers, 0.3% have no formal education or did not complete primary school, 4% had completed primary school while 44.2% had completed secondary school. Among respondents' mothers, only 1 did not undergo formal education, 2 did not complete primary school (0.6%), 14 had completed primary school (4.0%), and 138 had completed secondary school (39.1%).

Most of the respondents' parents are working as government employees (40.8% of fathers; 39.7% of mothers). Among the respondents' fathers, this is followed by a lower percentage of them, 32.6%, who are self-employed, 20.4% working at private sectors, 4.5% are already retired, and 1.7% are unemployed. Among the respondents' mothers, the second highest percentage, 24.6%, of them are self-employed followed by 20.4% who are unemployed while 13.6% are working at private sectors and the least are those who had retired (1.7%). Regarding marital status of respondents' parents, 6.8% of them are divorced, while the majority, 93.2%, are still married.

Therefore, in term of demographic characteristics, the respondents comprised of mostly females; lower secondary students; had parents that completed college or university education; had parents who are government employees; and had parents who are still married.

Table 1. Demographic characteristics of students (N = 353)

Variable	**n**	**(%)**
Gender		
Male	133	37.7
Female	220	62.3
Form		
One	111	31.4
Two	106	30.0
Three	79	22.4
Four	33	9.3
Five	20	5.7
Six	4	1.1
Father’s education		
No formal education	1	0.3
Did not complete primary school	1	0.3
Completed primary school	10	2.8
Completed secondary school	156	44.2
Completed college/university education	185	52.4
Mother’s education		
No formal education	1	0.3
Did not complete primary school	2	0.6
Completed primary school	14	4.0
Completed secondary school	138	39.1
Completed college/university education	198	56.1
Father’s occupation sector		
Government	144	40.8
Private	72	20.4
Self-employed	115	32.6
Retired	16	4.5
Unemployed	6	1.7
Mother’s occupation sector		
Government	140	39.7
Private	48	13.6
Self-employed	87	24.6
Retired	6	1.7
Unemployed	72	20.4
Parents’ marital status		
Married	329	93.2
Divorced	24	6.8

5.2 THE LEVEL OF PHYSICAL ACTIVITY, HISTORY OF BEING BULLIED, SELF-ESTEEM LEVEL AND PREVALENCE OF DEPRESSION AND SUICIDAL IDEATION AMONG RESPONDENTS

Table 2. Physical activity, history of being bullied, self-esteem, depression and suicidal ideation among students

Variable	N	(%)
Physical activity		
Low	123	34.8
Moderate	159	45.0
High	71	20.1
Being bullied at school		
Yes	133	37.7
No	220	62.3
Being bullied on electronic media		
Yes	34	9.6
No	319	90.4
Self-esteem level		
Low	93	26.3
Moderate	228	64.6
High	32	9.1
Depression		
Yes	181	51.3
No	172	48.7
Suicidal ideation		
Yes	119	33.7
No	234	66.3

The level of physical activity was reported in categories (low activity levels, moderate activity levels or high activity levels). First, we need to calculate Metabolic Equivalent (MET) minutes a week by multiplying the respective MET value according to type of activities (walking = 3.3, moderate activity = 4, vigorous activity = 8) by the minutes the activity was carried out and again by the number of days that that activity was undertaken. For example, if someone reports walking for 30 minutes 5 days a week then the total MET minutes for that activity are 3.3 X 30 X 5=495 Met minutes a week. Then, we need to add the MET minutes achieved in each category (walking, moderate activity and vigorous activity) to get total MET minutes of physical activity a week. Those who had achieved a minimum total physical activity of at least 3000 MET minutes a week were categorized as having high physical activity level, while those with at least 600 MET minutes a week were regarded as having moderate physical activity level. Respondents that scored lower than that were categorized under low level of physical activity.

As for self-esteem level, the total scores were summed up and respondents were categorized as having low self-esteem if their total scores were 14 and below; 15 to 39 were categorized as moderate; while 40 and above were regarded as having high self-esteem. For depression status, respondents were categorized as having depression if their total scores were 10 and above, while those who scored 9 and below were regarded as not having depression.

Based on Table 2, most of the respondents were involved in moderate physical activity (45.0%), while only 20.1% have high physical activity level. Slightly more than one-third were bullied at school while 9.6% were bullied on electronic media. Most of them, 64.6%, had moderate level of self-esteem, while 26.3% had low self-esteem. Slightly over half of them, 51.3%, were screened to have depression, and about one third had suicidal ideation.

5.3 Depression and its associated factors among respondents

Table 3. Sociodemographic characteristics and the percentage of depression

Variable	**n**	**Depression**		***P* value**
		Yes (%)	**No (%)**	
Gender				0.000
Male	133	50 (37.6)	83 (62.4)	
Female	220	131 (59.5)	89 (40.5)	
Form				0.511
One	111	57 (51.4)	54 (48.6)	
Two	106	61 (57.5)	45 (42.5)	
Three	79	37 (46.8)	42 (53.2)	
Four	33	13 (39.4)	20 (60.6)	
Five	20	11 (55.0)	9 (45.0)	
Six	4	2 (50.0)	2 (50.0)	

Father’s education				0.880
Primary education and below	12	7 (58.3)	5 (41.7)	
Secondary education	156	80 (51.3)	76 (48.7)	
Tertiary education	185	94 (50.8)	91 (49.2)	
Mother’s education				0.205
Primary education and below	17	11 (64.7)	6 (35.3)	
Secondary education	138	76 (55.1)	62 (44.9)	
Tertiary education	198	94 (47.5)	104 (52.5)	
Father’s occupation sector				0.992
Government	144	75 (52.1)	69 (47.9)	
Private	72	36 (50.0)	36 (50.0)	
Self-employed	115	59 (51.3)	56 (48.7)	
Retired or unemployed	22	11 (50.0)	11 (50.0)	
Mother’s occupation sector				0.972
Government	140	71 (50.7)	69 (49.3)	
Private	48	26 (54.2)	22 (45.8)	
Self-employed	87	45 (51.7)	42 (48.3)	
Retired or unemployed	78	39 (50.0)	39 (50.0)	

Table 4. Physical activity, history of being bullied, self-esteem and the percentage of depression

Variable	**n**	**Depression**		***P* value**
		Yes (%)	**No (%)**	
Physical activity				0.459
Low	123	60 (48.8)	63 (51.2)	
Moderate	159	80 (50.3)	79 (49.7)	
High	71	41 (57.7)	30 (42.3)	
Being bullied at school				0.000
Yes	133	86 (64.7)	47 (35.3)	
No	220	95 (43.2)	125 (56.8)	
Being bullied on electronic media				0.354
Yes	34	20 (58.8)	14 (41.2)	
No	319	161 (50.5)	158 (49.5)	
Self-esteem level				0.000
Low	93	75 (80.6)	18 (19.4)	
Moderate	228	101 (44.3)	127 (55.7)	
High	32	5 (15.6)	27 (84.4)	

Depression and its associated factors among the respondents were shown in Table 3. There was a higher prevalence of depression among females, 59.5%, as compared to males (37.6%). Form two students have the highest depression prevalence at 57.5%, while form four students had the lowest (39.4%). Respondents with fathers that had primary education or lower had the highest prevalence of depression, 58.3%, while those whose fathers had secondary and tertiary education had 51.3% and 50.8% prevalence respectively. Among the respondents, the highest prevalence of depression is in those whose mothers had primary education or lower (64.7%). This is followed by respondents whose mothers had secondary (55.1%) and tertiary education (47.5%). In term of parents' occupation sectors, the prevalence of depression is more or less the same among each respondents' group. Those whose fathers are government employees have 52.1% depression prevalence, followed by self-employed at 51.3%, while those in the private sector as well as retired/unemployed group had 50.0% prevalence each respectively. However, respondents with mothers working in private sectors have the highest depression prevalence (54.2%), followed by those with self-employed mothers (51.7%), government employee mothers (50.7%), and retired or unemployed mothers (50.0%). Respondents whose parents have divorced had 70.8% prevalence of depression, as compared to those whose parents are still together (49.8%).

In term of physical activity level, a high activity level is associated with higher prevalence of depression (57.7%), followed by moderate level (50.3%), and low level (48.8%). In addition, respondents with history of being bullied at school or on electronic media have a higher depression prevalence (64.7% and 58.8%, respectively) as compared to their colleagues who do not have these characteristics. More respondents with low self-esteem level were depressed (80.6%) compared to those with moderate (44.3%) and high self-esteem level (15.6%).

Depression was significantly associated with gender, parents' marital status, history of being bullied at school, and self-esteem level. However, following binary logistic regression (as shown in Table 4), depression was significantly associated with female gender (adjusted odds ratio (aOR): 2.141; 95% CI 1.303 to 3.516; p = 0.003), history of being bullied at school (aOR: 2.346; 95% CI 1.430 to 3.848; p = 0.001), low self-esteem level (aOR: 16.291; 95% CI 5.326 to 49.829; p = 0.000), and moderate self-esteem level (aOR: 4.220; 95% CI 2.329 to 7.647; p = 0.000). The binary logistic regression showed an overall fit model with Hosmer and Lemeshow test (p = 0.316).

Table 5. The risk of depression among respondents

Variables	**Total**	**Depression**		**Adjusted OR**	**95% CI**	**p-value**
		n	**%**			
Gender						
Male	133	50	37.6	1		
Female	220	131	59.5	2.141	1.303 – 3.516	0.003
Parents' marital status						
Married	329	164	49.8	1		
Divorced	24	17	70.8	2.501	0.909 – 6.881	0.076
Being bullied at school						
Yes	133	86	64.7	2.346	1.430 – 3.848	0.001
No	220	95	43.2	1		
Self-esteem level						
Low	93	75	80.6	16.291	5.326 – 49.829	0.000
Moderate	228	101	44.3	4.220	2.329 – 7.647	0.000
High	32	5	15.6	1		

Females had approximately 2.1 times higher risk of developing depression than males (aOR: 2.141; 95% CI 1.303 to 3.516; p = 0.003). Similarly, respondents who had been bullied at school were about 2.3 times higher risk of depression as compared to those who were not bullied (aOR: 2.346; 95% CI 1.430 to 3.848; p = 0.001). Based on self-esteem level, respondents who were having low self-esteem are 16.3 times higher risk in developing depression (aOR: 16.291; 95% CI 5.326 to 49.829; p = 0.000). Again, when we compare with the high self-esteem group, the moderate group on the other hand, had about 4.2 times higher risk to be depressed (aOR: 4.220; 95% CI 2.329 to 7.647; p = 0.000).

Overall, there is no statistically significant association between depression and students' form levels; parents' level of education, occupation sectors, parents' marital status; level of physical activity; and history of being bullied on electronic media. After adjusting for the significant associated factors, the strongest predictor of depression among the respondents was having low self-esteem level, which had the highest adjusted odds ratio of 16.291.

5.4 ASSOCIATION BETWEEN DEPRESSION AND SUICIDAL IDEATION

Table 6 shows that the overall prevalence of suicidal ideation among respondents is 33.7%.

Table 6. Prevalence of suicidal ideation among the respondents

	Frequency (N = 353)	**Prevalence (%)**
Suicidal ideation		
Yes	119	33.7
No	234	66.3

Based on Table 7, the prevalence of suicidal ideation among respondents with depression is 54.1%. It also shows that they have about 8.5 times higher risk of having suicidal ideation (OR: 8.490; 95% CI 4.937 – 14.599; p = 0.000) as compared to those who were not depressed.

Table 7. Association between depression and suicidal ideation

Variable	**Total**	**Suicidal Ideation**		**Odds Ratio**	**95% CI**	**p-value**
		N	**%**			
Depression						
Yes	181	98	54.1	8.490	4.937 – 14.599	0.000
No	172	21	12.2	1		

CHAPTER 5

DISCUSSION

The prevalence of depression in this study was found to be 51.3%, which represents 181 out of 353 respondents. This prevalence is higher as compared to a study done by Ibrahim et al. (2017) among secondary school students, which was 32.7%. It is also higher as compared to the results of other studies done by Wahab et al. (2013) and Yaacob et al. (2009), whereby the prevalence of depression among secondary school students was 39.7% and 24.2%, respectively. It is important to note that the two studies mentioned above used Patient Health Questionnaire-9 (PHQ-9) and Children Depression Inventory (CDI) questionnaires. Wahab et al. (2013) used the same questionnaire with this present study, which is the validated Malay version of 21-item Depression, Anxiety, Stress Scale (DASS-21).

The reason for the high prevalence of depression could be due to the fact that the respondents were from religious schools with a specially developed syllabus with busy schedules, as mentioned in the methodology. This packed schedule with little time to rest or sleep may be one of the factors that give rise to the high prevalence of depression among the respondents. The result is consistent with several previous studies. Roberts and Duong (2014) found that having daily sleep of 6 hours or less may be a precursor for major depression in adolescents aged 11 to 17 years. Another study done by Fredriksen et al. (2004) found that adolescents aged 11 to 14 years who obtained less sleep over time experienced heightened level of depressive symptoms.

Based on the research results, there are three statistically significant factors associated with depression. These are gender, history of being bullied at school, and self-esteem level. This study reveals that there is significant association between gender and depression, in which females have a higher risk of being depressed as compared to males (aOR: 2.141; 95% CI 1.303 to 3.516; $p = 0.003$). In term of prevalence between different genders, female has higher prevalence of depression (59.5%) as opposed to male (37.6%).

As shown in a study by Faravelli et al. (2013), they found that females had 2.2 times higher risk than males to be depressed, and the incidence of depression in females aged 14 to 19 years is 85.7%. In addition, the result of this study may support the gonadic theory, in which Oldehinkel and Bouma (2011) proposed the possible hypothalamus-pituitary-adrenal cortex (HPA) axis role in modulating mood and behavior, that results in increased females' sensitivity reaction towards stressful life events as compared to males. It is suggested that estrogen withdrawal during menstruation may interfere with the ability to neutralize glucocorticoids released during stress, therefore increasing the vulnerability for depression.

From this study, it was found that those who were victims of bullying at school were at 2.3 times higher risk of developing depression as compared to those who were not (aOR: 2.346; 95% CI 1.430 to 3.848; $p = 0.001$). A study done by Kaltiala-heino and Frojd (2011) found that adolescence is the time where peer relationships are of utmost importance, therefore traumatic peer-related experience such as being bullied can be severe enough to cause depression. In addition, the victims of bullying are often characterized as highly malleable with helpless behavior, and are less popular among their peers. All of the characteristics mentioned above not only can be a predisposing factor of being bullied at school, but also act as antecedents of depression itself.

In a cohort study by Bowes et al. (2015), the researchers found that increasing frequency of being bullied corresponded with increasing prevalence of depression among adolescents. Although in this research the author did not measure the frequency of being bullied among the respondents, it is suggested that being bullied repeatedly at school may be one of the contributing factors of higher depression prevalence among the respondents.

Self-esteem is part of the elements that make up an individual's self-concept, whereby it is related to ones' appraisal of their positive and negative value, or how they estimate their self-value in the real life (Harter, 1999). In this study, respondents who are having low self-esteem were 16.3 times at higher risk of developing depression as compared to those with high self-esteem. This is supported by a recent study by Fiorilli et al. (2019) which stated that the dysfunctional way to manage incompetent and worthless feeling among low self-esteem adolescents will lead to stress and finally, depression. In another study by Ibrahim et al. (2017), the researchers found that low self-esteem had significant association with depression among adolescents. They concluded that those with low self-esteem had a 3 times higher risk of developing depression than those with high self-esteem. The self-esteem dimensions being mentioned in Ibrahim et al.'s study, which include poor interpersonal relationships; unable to control life events; and negative emotions management, showed the highest correlation with depression. Fiorilli et al. (2019) suggested that the higher the self-esteem level, the lower was the risk of being depressed. As compared to high self-esteem group, respondents in this study with moderate self-esteem level had 4.2 times risk of depression, that is much lower than low self-esteem respondents who had a 16.3 times risk.

In this study, there is a 33.7% prevalence of suicidal ideation among the respondents overall. This is higher as compared to a study by Ibrahim et al. (2017), that showed a suicidal ideation prevalence of 27.9% among adolescents aged 13 to 17 years. The same type of questionnaire was used (PHQ-9), therefore the results may be regarded as comparable for both studies. The finding of this study was also higher than several other studies among Malaysian school-going adolescents, which were 6.2% (Chan et al., 2016), 7.9% (Ahmad et al., 2014), and 7.0% (Chen et al., 2005).

When we take a deeper look, the prevalence of suicidal ideation among depressed respondents is 54.1%, or 8.5 times higher than those who were not depressed (12.2%). The findings are in line with the study by Ibrahim et al. (2017), which stated that depression was one of the predictors of suicidal ideation, with 55.8% prevalence among depressed respondents as compared to those who were not depressed (14.4%). In another study, Talib and Abdollahi (2015) found a significant relationship between depression and suicidal behavior. Harris and Lennings (1993) also found depression to be a significant predictor for suicide in adolescents. Overall, we can say that while depression is associated with suicidal ideation, it can lead to a more serious problem of suicidal behavior if not managed properly.

There are a few limitations in the conduct of this study. One is the type of study conducted which is the cross-sectional study design. In this study, the researcher is only able to find the prevalence of depression. However, he could not prove any causal relationship between the associated factors and depression. This type of study design also could not determine whether the outcome followed the exposure or vice versa. Second, the study was only conducted among school-going adolescents of private secondary boarding schools in Ipoh and Kuala Kangsar, Perak. Thus, the results cannot be generalized to the general adolescents' population in Perak, or Malaysia as a whole. Next, some of the questions being asked, such as in DASS-21 and IPAQ may introduce recall bias. This is due to the fact that respondents were required to recall their feelings and amount of time they had spent on certain things for the past seven days, which can lead to inaccurate information being provided. Last but not least, the author did not assess the amount of sleeping hours of respondents, which may be one of the contributing factors of depression as mentioned earlier. It is hope that any researcher who shall conduct a similar study in the future, will not neglect this important risk factor as part of the variables being studied.

CHAPTER 6

CONCLUSION AND RECOMMENDATION

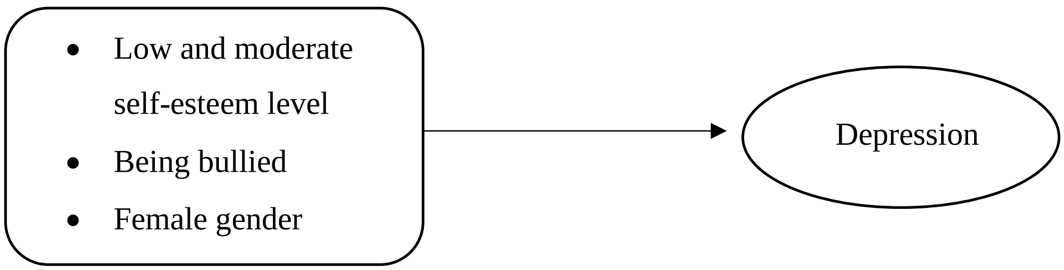

Figure 3: Significant associated factors of depression

The purpose of the study was to investigate the prevalence and associated factors of depression among adolescents in religious-based secondary boarding schools in Ipoh and Kuala Kangsar, Perak. The findings showed that the prevalence of depression among the respondents is alarmingly high as compared to other studies. It is important to note that the instruments utilized in this study (like DASS-21) were for screening purpose rather than diagnostic. Therefore, these school-going adolescents need to be further assessed accordingly with diagnostic instruments, so that proper management can be carried out. Early detection and treatment of depression is crucial, to prevent unwanted and dangerous complications such as suicide. Based on this study, the most prominent risk factor for depression was having low level of self-esteem, with the highest adjusted odds ratio of 16.291. This was followed by moderate self-esteem level, being bullied at the school, and female gender.

The researcher proposes that mental health education programs with depression as its primary focus, are urgently needed to be carried out for secondary school students. The health education can be in the form of talk, group discussion, campaign, flyers distribution and others. The importance of health education is mainly to promote healthy attitude and coping strategies towards stressful life events, good interpersonal relationship, and self-esteem improvement strategies. An effort to educate the people comprising the management, teachers, students and parents on issue of depression and suicidal occurring among religious school students should also be carried out. Healthcare professionals as well as religious authorities must work together to disseminate depression awareness and correct a rather common mindset, that religion alone can prevent someone from having depression or being suicidal. All other associated factors like gender, low self-esteem, being bullied, parental divorce and others, should be taken into consideration.

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APPENDICES

Appendix A: Questionnaires

No. Kad Pengenalan | | | | | | | | | | | | |

UNIVERSITI KUALA LUMPUR
ROYAL COLLEGE OF MEDICINE PERAK

BORANG SOAL SELIDIK

FAKTOR-FAKTOR YANG BERKAITAN DENGAN KEMURUNGAN DALAM KALANGAN REMAJA DI SEKOLAH MENENGAH BERASRAMA BERTERASKAN AGAMA DI IPOH DAN KUALA KANGSAR, PERAK

Encik/Cik,
Assalamualaikum dan terima kasih kerana bersetuju untuk mengambil bahagian dalam kaji selidik ini. Saya pelajar Sarjana Sains Kesihatan Awam daripada UniKL RCMP. Saya sedang menjalankan kajian mengenai faktor-faktor yang berkaitan dengan kemurungan dalam kalangan remaja. Jadi, saya mengharapkan bantuan anda agar dapat menjawab beberapa soalan. Keputusan yang akan diperolehi di sini adalah semata-mata untuk tujuan Pendidikan, dan dalam apa jua keadaan tidak akan digunakan untuk mengaut keuntungan sama ada dibenarkan ataupun tidak. Soal selidik ini hanya mengambil masa lebih kurang 20 minit. Terima kasih.

<u>BAHAGIAN A: BIODATA</u>

No.	**Soalan**	**Jawapan**		**Untuk penyelidik sahaja**
Q1	Umur lengkap pada tahun ini	[ ][ ] tahun		
Q2	Kewarganegaraan	Malaysia		
		Bukan Malaysia		
Q3	Jantina	Lelaki		

		Perempuan		
Q4	Tingkatan	Tingkatan satu		
		Tingkatan dua		
		Tingkatan tiga		
		Tingkatan empat		
		Tingkatan lima		
Q5	Tahap pendidikan tertinggi bapa	Tiada pendidikan formal		
		Tidak menghabiskan pendidikan sekolah rendah		
		Tamat pendidikan sekolah rendah		
		Tamat pendidikan sekolah menengah		
		Tamat pendidikan di kolej/universiti		
Q6	Tahap pendidikan tertinggi ibu	Tiada pendidikan formal		
		Tidak menghabiskan pendidikan sekolah rendah		
		Tamat pendidikan sekolah rendah		
		Tamat pendidikan sekolah menengah		
		Tamat pendidikan di kolej/universiti		
Q7	Sektor pekerjaan bapa	Sektor kerajaan		
		Sektor swasta		
		Bekerja sendiri		
		Pesara		
		Tidak bekerja		
Q8	Sektor pekerjaan ibu	Sektor kerajaan		
		Sektor swasta		
		Bekerja sendiri		
		Pesara		
		Tidak bekerja		
Q9	Anggaran jumlah pendapatan bulanan isi rumah	RM______________		
Q10	Status perkahwinan ibu bapa	Berkahwin		
		Sudah bercerai		

BAHAGIAN B: AKTIVITI FIZIKAL

Soalan-soalan berikut adalah tentang jumlah masa yang anda gunakan untuk **aktif secara fizikal dalam tempoh 7 hari yang lepas**.

No.	Soalan	Jawapan	Untuk penyelidik sahaja
Q1	Adakah anda melakukan apa-apa **aktiviti berat** dalam 7 hari yang lepas? Aktiviti berat ialah aktiviti yang menyebabkan anda **bernafas jauh lebih kuat daripada biasa,** yang anda lakukan sekurang-kurangnya 10 minit pada sesuatu masa.	*Ya* ☐ *Tidak* ☐ **Jika jawapan anda adalah* ***tidak****, sila langkau ke Q4. Tidak perlu jawab soalan Q2 dan Q3.*	
Q2	Dalam tempoh 7 hari yang lepas ini, **berapa hari** anda telah melakukan aktiviti fizikal berat, contohnya mengangkat barang berat, mencangkul, senaman aerobik atau berbasikal laju?	☐ hari	
Q3	Berapakah **masa** yang anda biasa gunakan untuk melakukan aktiviti fizikal berat pada **salah satu daripada hari berkenaan**? (Jumlah masa anda melakukan aktiviti tersebut dalam sehari) *Contoh: 1 jam 20 minit*	☐ jam ☐ minit	
Q4	Adakah anda melakukan apa-apa **aktiviti sederhana** dalam 7 hari yang lepas? Aktiviti sederhana adalah aktiviti yang menyebabkan anda bernafas **agak kuat** daripada biasa, yang anda lakukan sekurang-kurangnya 10 minit pada sesuatu masa.	*Ya* ☐ *Tidak* ☐ **Jika jawapan anda adalah* ***tidak****, langkau ke Q7. Tidak perlu jawab soalan Q5 dan Q6.*	
Q5	Dalam tempoh 7 hari yang lepas ini **berapa hari** anda telah melakukan aktiviti fizikal sederhana, contohnya mengangkat muatan ringan, mengelap lantai, berbasikal pada kelajuan biasa, atau bermain	☐ hari	

	badminton beregu? Ini tidak termasuk berjalan kaki.		
Q6	Berapakah **masa** yang anda biasa gunakan untuk melakukan aktiviti fizikal sederhana pada **salah satu daripada hari berkenaan**? (Jumlah masa anda melakukan aktiviti tersebut dalam sehari) *Contoh: 1 jam 20 minit*	☐ jam ☐ minit	
Q7	Dalam tempoh 7 hari yang lepas ini, **berapa hari** anda telah berjalan kaki selama sekurang-kurangnya 10 minit pada sesuatu masa? Ini merangkumi berjalan kaki di sekolah dan di rumah.	☐ hari **Jika jawapannya adalah kosong (0) hari, langkau ke Q9. Tidak perlu jawab soalan Q8.*	
Q8	Berapakah **masa** yang anda biasa gunakan untuk berjalan kaki pada **salah satu daripada hari berkenaan**? (Jumlah masa anda berjalan kaki dalam sehari) *Contoh: 1 jam 20 minit*	☐ jam ☐ minit	
Q9	Dalam tempoh 7 hari yang lepas ini, berapakah **jumlah masa** yang anda telah gunakan untuk duduk pada sesuatu hari? *Contoh: 1 jam 20 minit*	☐ jam ☐ minit	

<u>BAHAGIAN C: SOALAN-SOALAN BERKENAAN BULI</u>

Buli ialah apabila seorang atau lebih pelajar mengejek, mengancam, menyebarkan khabar angin tentang seseorang, memukul, menolak, atau menyakiti pelajar lain **berulang kali**. Ia tidak dikatakan sebagai membuli apabila 2 orang pelajar yang mempunyai kekuatan atau kuasa yang sama berhujah, bertarung atau mengejek antara satu sama lain dengan cara yang mesra. Sila tandakan ✓ pada jawapan anda.

No.	Soalan	Jawapan	
1.	Pernahkah anda dibuli di dalam kawasan sekolah?	Ya	
		Tidak	
2.	Pernahkah anda dibuli secara elektronik? (Dibuli melalui mesej, Instagram, Facebook, atau media sosial lain)	Ya	
		Tidak	

BAHAGIAN D: SOALAN BERKAITAN PERSEPSI TERHADAP HARGA DIRI

Sila tandakan ✓ pada jawapan anda.

No.	Kenyataan-kenyataan	Sangat Tidak Setuju 1	Tidak Setuju 2	Berkecuali 3	Setuju 4	Sangat Setuju 5
1.	Saya selalu berpuas hati dengan diri saya					
2.	Adakalanya saya rasa diri saya ini tidak berguna langsung					
3.	Saya bersikap positif terhadap diri					
4.	Saya selalu rasa yang saya ini seorang yang gagal					
5.	Saya rasa yang saya ada beberapa kualiti yang baik					
6.	Saya rasa saya mempunyai nilai diri, sekurang-kurangnya sama seperti orang lain					
7.	Hajat saya ialah agar saya lebih menghormati diri saya					
8.	Kadangkala saya terfikir yang saya bukannya selalu baik					
9.	Saya boleh melakukan tugas sama baik seperti orang lain					
10.	Saya rasa tidak banyak yang boleh saya banggakan tentang diri saya					

BAHAGIAN E: SOAL SELIDIK KESIHATAN (PHQ-9)

Sila **bulatkan** nombor 0,1,2 atau 3 yang menggambarkan anda **sepanjang 2 minggu yang lalu**.

No.	Keadaan anda sepanjang 2 minggu yang lalu	Tiada langsung	Beberapa hari	Lebih daripada 7 hari	Hampir setiap hari
1.	Kurang berminat atau keseronokan dalam melakukan sesuatu perkara	0	1	2	3
2.	Rasa sedih, tidak gembira atau putus asa	0	1	2	3
3.	Masalah untuk tidur, atau tidak tidur nyenyak, atau tidur berlebihan	0	1	2	3
4.	Rasa letih atau mempunyai sedikit tenaga	0	1	2	3

5.	Kurang selera atau makan berlebihan	0	1	2	3
6.	Rasa buruk mengenai diri – seperti anda seorang yang gagal, atau anda telah menyebabkan diri anda atau keluarga anda kecewa	0	1	2	3
7.	Masalah untuk menumpukan perhatian ke atas sesuatu perkara	0	1	2	3
8.	Bergerak atau bercakap terlalu perlahan sehinggakan orang lain perasan, atau merasa sangat resah atau gelisah sehinggakan anda telah bergerak dengan lebih banyak daripada biasa	0	1	2	3
9.	Memikirkan adalah lebih baik jika anda mati sahaja, atau mencederakan diri sendiri	0	1	2	3

BAHAGIAN F: SOALAN BERKAITAN MASALAH KEMURUNGAN (DASS 21)

Sila **bulatkan** pada nombor 0,1,2 atau 3 bagi menggambarkan keadaan anda **sepanjang seminggu yang lalu**. Jangan ambil masa yang terlalu lama untuk menjawab mana-mana kenyataan.

Skala pemarkahan yang boleh anda pilih adalah seperti berikut:

0 *Tidak langsung menggambarkan keadaan saya*
1 *Sedikit atau jarang-jarang menggambarkan keadaan saya.*
2 *Banyak atau kerap kali menggambarkan keadaan saya.*
3 *Sangat banyak atau sangat kerap menggambarkan keadaan saya*

1. Saya dapati diri saya sukar ditenteramkan **0 1 2 3**

---

2. Saya sedar mulut saya terasa kering **0 1 2 3**

---

3. Saya langsung tidak dapat mengalami perasaan positif **0 1 2 3**

---

4. Saya mengalami kesukaran bernafas (contohnya pernafasan yang laju, tercungap-cungap walaupun tidak melakukan senaman fizikal) **0 1 2 3**

---

5. Saya sukar untuk mendapatkan semangat bagi melakukan sesuatu perkara **0 1 2 3**

---

6. Saya suka untuk bertindak keterlaluan dalam sesuatu keadaan **0 1 2 3**

---

7. Saya rasa menggeletar (contohnya pada tangan) **0 1 2 3**

8. Saya rasa saya menggunakan banyak tenaga dalam keadaan cemas	**0 1 2 3**
9. Saya bimbang keadaan di mana saya mungkin menjadi panik dan melakukan perkara yang membodohkan diri sendiri	**0 1 2 3**
10. Saya rasa saya tidak mempunyai apa-apa untuk diharapkan	**0 1 2 3**
11. Saya dapati diri saya semakin gelisah	**0 1 2 3**
12. Saya rasa sukar untuk relaks	**0 1 2 3**
13. Saya rasa sedih dan murung	**0 1 2 3**
14. Saya tidak dapat menahan sabar dengan perkara yang menghalang saya daripada meneruskan apa yang saya lakukan	**0 1 2 3**
15. Saya rasa hampir-hampir menjadi panik/cemas	**0 1 2 3**
16. Saya tidak bersemangat dengan apa jua yang saya lakukan	**0 1 2 3**
17. Saya seorang yang tidak begitu berharga	**0 1 2 3**
18. Saya rasa yang saya mudah tersentuh	**0 1 2 3**
19. Saya sedar tindak balas jantung saya walaupun tidak melakukan aktiviti fizikal (contohnya kadar denyutan jantung bertambah, atau denyutan jantung berkurangan)	**0 1 2 3**
20. Saya berasa takut tanpa sebab yang munasabah	**0 1 2 3**
21. Saya rasa hidup ini tidak bermakna	**0 1 2 3**

-SOAL SELIDIK TAMAT-

Appendix B: Scoring Guidelines For Questionnaires

1) International Physical Activity Questionnaire (IPAQ)

There are two forms of output from scoring the IPAQ. Results can be reported in categories (low activity levels, moderate activity levels or high activity levels) or as a continuous variable (MET minutes a week). MET minutes represent the amount of energy expended carrying out physical activity.

To get a continuous variable score from the IPAQ (MET minutes a week) we will consider walking to be 3.3 METS, moderate physical activity to be 4 METS and vigorous physical activity to be 8 METS.

Physical activity categories:

Those who score **HIGH** on the IPAQ engage in

- Vigorous intensity activity on at least 3 days achieving a minimum total physical activity of at least 1500 MET minutes a week

OR

- 7 or more days of any combination of walking, moderate intensity or vigorous intensity activities achieving a minimum total physical activity of at least 3000 MET minutes a week.

Those who score **MODERATE** on the IPAQ engage in

- 5 or more days of any combination of walking, moderate intensity or vigorous intensity activities achieving a minimum total physical activity of at least 600 MET minutes a week.

Scoring a **LOW** level of physical activity on the IPAQ means that you are not meeting any of the criteria for either MODERATE of HIGH levels of physical activity.

Calculating results:

- Bouts of activity lasting less than 10 minutes duration are not counted.
- Convert all activity to minutes before calculating MET minutes. Doing the math in hours will give you incorrect results.
- It is recommended that activity bouts of greater than 3 hours are truncated. That is to say that a bout cannot be longer than 3 hours (180 minutes). This means that in each category a maximum of 21 hours of activity are permitted a week (3 hours X 7 days)
- To calculate MET minutes a week multiply the MET value given (remember walking = 3.3, moderate activity = 4, vigorous activity = 8) by the minutes the activity was carried out and again by the number of days that that activity was undertaken. For example, if someone reports walking for 30 minutes 5 days a week then the total MET minutes for that activity are 3.3 X 30 X 5=495 Met minutes a week.

- You can add the MET minutes achieved in each category (walking, moderate activity and vigorous activity) to get total MET minutes of physical activity a week.

2) Rosenberg Self-esteem Scale (RSES)

It consists of 10 questions, with scores ranging from 1 to 4. Questions 1,3,5,6 and 9 are positively directed questions, therefore they are scored by adding all the scored selected by the respondents. Questions 2,4,7 and 10 are negatively directed questions. For these questions, reverse analysis was used; whereby scores of 1 (strongly agree), 2 (agree), 3 (neither agree nor disagree), 4 (disagree), and 5 (strongly disagree) were used.

The total scores were then summed up and respondents were categorized as having low self-esteem if their total scores were 14 and below. 15 to 39 were categorized as moderate, while 40 and above were regarded as having high self-esteem.

3) Depression Anxiety and Stress Scale (DASS-21)

It is a set of three self-report scales designed to measure the emotional states of depression, anxiety and stress. Each of the three DASS-21 scales contains 7 items, divided into subscales with similar content. The depression scale assesses dysphoria, hopelessness, devaluation of life, self-deprecation, lack of interest / involvement, anhedonia and inertia.

The anxiety scale assesses autonomic arousal, skeletal muscle effects, situational anxiety, and subjective experience of anxious affect. The stress scale is sensitive to levels of chronic non-specific arousal. It assesses difficulty relaxing, nervous arousal, and being easily upset / agitated, irritable / over-reactive and impatient.

The questions that are referring to depression, are questions number 3,5,10,13,16,17 and 21. Scores on the DASS-21 will need to be multiplied by 2 to calculate the final score.

The DASS-21 has no direct implications for the allocation of patients to discrete diagnostic categories postulated in classificatory systems such as the DSM and ICD. Recommended cut-off scores for conventional severity labels (normal, moderate, severe) are as follows:

	Scores
Normal	0-9
Mild	10-13
Moderate	14-20
Severe	21-27
Extremely severe	28+

Appendix C: Letter Of Ethical Approval

UNIVERSITI KUALA LUMPUR
ROYAL COLLEGE OF MEDICINE PERAK
No. 3, Jalan Greentown,
30450 Ipoh, Perak Darul Ridzuan.
Tel : (605) 243 2635
Fax : (605) 243 2636
Website : www.rcmp.unikl.edu.my

UniKLRCMP/MREC/2019/FTMSPH-006

18th July 2019

Muhammadbazli Aminmokhsin bin Aminuddin
Student Master of Science In Public Health
Year 1 (Jan/2019)
Faculty of Medicine
UniKL RCMP

Dear All,

MEDICAL RESEARCH ETHICS COMMITTEE: APPROVAL FOR STUDENT RESEARCH PROJECT

The above matter refers.

Title of Research: **Factors Associated With Depression Among Adolescents In Religious-Based Secondary Schools In Ipoh and Kuala Kangsar, Perak.**

We are pleased to inform that the research proposal submitted has been approved via the expedited review by the MREC committee.

The research is reviewed via the expedited-review process are minimal risk to the participants and meet one or more of the Expedited-Review Criteria.

On behalf of the committee, I wish the best in your future researches.

"Where Knowledge Is Applied"

Yours sincerely,

PROFESSOR DR OSMAN ALI
Chairman
Medical Research Ethics Committee (MREC)

c.c Dr. Rosalind Simon, MREC Committee
Mdm Lew Choi Fong, MREC Committee
Prof. Dr. Gouri Kumar Dash, MREC Committee
Assoc. Prof. Dr. Fatehpal Malhi a/l Waryam Singh, MREC Committee
Prof. Dato' Dr Wahinuddin Bin Sulaiman, MREC Committee

oa/bohb

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Managed by: P.C.M. Sdn. Bhd. (477486-U)
A Subsidiary of Universiti Teknikal MARA Sdn. Bhd. (570132-T)

Group of Companies

Appendix D: Permission To Conduct The Study

Muhammadbazli Aminmokhsin Bin Aminuddin
Pelajar Sarjana Sains Kesihatan Awam,
Universiti Kuala Lumpur RCMP,
Perak, Malaysia.

AF Ustaz Mohd Sabri Bin Nordin,
Ketua Pegawai Eksekutif,
Yayasan Pendidikan Maahad Tahfiz al Quran ADDIN Perak,
Tingkat 1, Kompleks Pendidikan Masjid Al Aulia',
Taman Sin Lok, Jalan Kuala Kangsar,
30010 Ipoh, Perak Darul Ridzuan.

29 Julai 2019

Tuan,

PERMOHONAN KEBENARAN MENJALANKAN KAJIAN PENYELIDIKAN DI SEKOLAH-SEKOLAH DI BAWAH YAYASAN ADDIN

Dengan segala hormatnya perkara di atas adalah dirujuk.

2. Sukacita dimaklumkan kepada tuan bahawa saya ialah pelajar Universiti Kuala Lumpur RCMP, Perak, ingin memohon kebenaran untuk menjalankan penyelidikan peringkat sarjana di institusi di bawah seliaan pihak tuan.

3. Maklumat permohonan saya adalah seperti berikut:

Nama	: Muhammadbazli Aminmokhsin Bin Aminuddin
No. KP	: 900106-08-5377
No. Tel	: 011-10132468
E-mel	: bez.aminuddin@gmail.com
Tajuk Kajian	: Faktor Yang Berkaitan Dengan Kemurungan Dalam Kalangan Remaja Di Sekolah Menengah Agama Di Ipoh Dan Kuala Kangsar, Perak.

5. Sehubungan dengan itu, saya memohon kepada tuan agar saya memperoleh kebenaran untuk menjalankan penyelidikan ini, demi meningkatkan lagi kualiti pendidikan negara khususnya di bidang Tahfiz Al-Quran. Segala perhatian dan keperihatinan pihak tuan diucapkan jutaan terima kasih.

Sekian.

DILULUSKAN

Yang benar,

....
(Muhammadbazli Aminmokhsin Bin Aminuddin)